IBM SkillsBuild Internship Project: Data Analytics & AI

📌 Project Overview

This repository contains the complete internship project submission for the **IBM SkillsBuild Data Analytics & AI Internship Program** organized by **BharatCares** in association with **AICTE**.
The objective of this project is to apply data cleaning, exploratory data analysis (EDA), statistical modeling, and data visualization techniques on business datasets to extract actionable operational insights.

👤 Author Details

- **Student Name:** Sachin
- **Program:** IBM SkillsBuild Data Analytics & AI Internship
- **Partner Organizations:** BharatCares & AICTE
- **Domain:** Data Analytics & AI

📁 Repository Structure & Deliverables

The repository consists of four key deliverables prepared for submission:

1. Sachin_DataAnalyticsProject.py - Core Python source script containing data generation, preprocessing, EDA, and statistical visualization pipelines.
2. requirements.txt - Complete list of Python packages and libraries required to execute the project.
3. Sachin_ProjectReport.md - Comprehensive Markdown project report outlining objectives, dataset descriptions, methodology, visual analysis, and business recommendations.
4. README.md - Documentation and project guide.

🛠️ Tech Stack & Dependencies

- **Programming Language:** Python 3.8+
- **Data Processing:** pandas, numpy
- **Data Visualization:** matplotlib, seaborn
- **Machine Learning & Analytics:** scikit-learn
- **IDE / Environment:** Jupyter Notebook / Visual Studio Code / Google Colab

🚀 How to Set Up and Run

1. Clone or Download

Clone this repository to your local computer or download all project files into a single directory.

2. Install Dependencies

Open your terminal or command prompt and run:
```
pip install -r requirements.txt
```

3. Run the Main Analytics Script

Execute the main script using:
```
python Sachin_DataAnalyticsProject.py
```

4. Output

Running the script will process the dataset and automatically generate and save the following visualization plots in your directory:

- income_distribution.png
- income_vs_spending.png
- correlation_heatmap.png

📊 Summary of Key Analysis

- **Data Quality:** Checked and validated clean structures across numeric and categorical variables.
- **Customer Clustering:** Analyzed relationships between annual income levels and customer spending scores across categories (Electronics, Clothing, Home, Books).
- **Insights:** Identified key high-value consumer groups to enable data-driven targeted marketing strategies.

Submitted by Sachin for IBM SkillsBuild Internship Evaluation.